

EEG Signal Detection Using Machine Learning

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Abstract

Electroencephalogram (EEG) signals provide a non-invasive and dynamic method for analyzing brain activity, playing a crucial role in emotion recognition systems. This report presents a detailed study on classifying emotional states using EEG data and machine learning techniques. Two supervised learning models—K-Nearest Neighbors (KNN) and Support Vector Machine (SVM)—were developed and evaluated using a high-dimensional EEG dataset. Experimental results demonstrate that both models are effective in classifying emotional states, with SVM achieving higher overall performance across multiple evaluation metrics.

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1. Introduction

Emotion recognition from EEG signals is an emerging area of research with vast applications in brain-computer interfaces (BCIs), mental health diagnostics, and affective computing. However, the inherently complex and high-dimensional nature of EEG data presents significant challenges in traditional statistical analysis. Machine learning provides powerful tools for identifying patterns and enabling robust emotion classification. This study investigates and compares the performance of KNN and SVM classifiers for detecting emotional states based on EEG signals.

2. Literature Survey

In [1], Mu et al. presented a comprehensive survey of machine learning methods for EEG-based mental health detection, emphasizing the importance of robust preprocessing and feature extraction. The review highlights the use of time-domain features (e.g., mean, standard deviation), frequency-domain features (e.g., alpha, beta, theta band power), and nonlinear dynamics (e.g., entropy, fractal dimension) in classifying mental health conditions. These features serve as critical inputs to machine learning algorithms like SVM, KNN, and decision trees, which have been widely adopted due to their interpretability and relatively low computational cost.

In [2], Darzi et al. applied wavelet-transformed EEG features for stress level classification using conventional classifiers. Their findings demonstrated that machine learning models, particularly Support Vector Machines (SVM), were capable of detecting stress states with high accuracy and generalization when trained on carefully extracted features. The study underlined the significance of preprocessing steps such as signal segmentation, normalization, and noise filtering.

In [3], Saeed et al. explored the detection of major depressive disorder using EEG signals and a variety of classifiers, including SVM. Their study reported classification accuracies close to 90%, using statistical and spectral features such as band power and Hjorth parameters. These results confirmed the viability of using traditional ML models for mental health prediction, especially in clinical environments where computational efficiency and model interpretability are important.

In [4], Sharma et al. conducted a meta-analysis of machine learning approaches used in EEG-based mental health applications. Their analysis reinforced that SVM consistently outperforms other shallow classifiers in terms of precision, recall, and F1-score across diverse mental health conditions, including anxiety, depression, and attention disorders. The authors emphasized the need for careful feature engineering and subject-specific normalization to deal with inter-subject variability in EEG signals.

In [5], Al-Shargie et al. demonstrated the integration of EEG with functional near-infrared spectroscopy (fNIRS) for improved stress assessment. Although not strictly limited to machine learning, their findings suggested that multimodal approaches could further enhance model robustness and reliability, an insight valuable for hybrid systems that fuse EEG with other biosignals in ML pipelines.

In [6], Craik et al. reviewed deep learning methods for EEG analysis, noting that although Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) deliver state-of-

the-art performance, traditional machine learning models like SVM remain competitive for smaller datasets with engineered features. The study stressed that deep learning approaches, while powerful, often suffer from a lack of interpretability and high training data demands.

In [7], Zheng and Lu investigated the role of frequency bands and EEG channels in emotion recognition tasks using SVM and deep neural networks. Their research emphasized that certain frequency bands, particularly theta and alpha, contain rich information for emotion classification, which directly supports the relevance of feature-based SVM models in affective computing.

In [8], Subhani et al. applied machine learning for the classification of depressive symptoms using EEG signals. They reported that combining nonlinear features with SVM achieved better performance than linear models or raw time-domain data. The study demonstrated that EEG signals are a viable modality for automatic mental health diagnostics when paired with appropriate ML techniques.

In [9], Sun et al. developed a hybrid model combining time-frequency domain features and SVM to recognize emotional states. Their classifier achieved significant performance improvements compared to conventional models, indicating the effectiveness of multi-feature fusion techniques in EEG-based detection systems. The success of their SVM model supports its inclusion in diagnostic pipelines for mood disorders.

Finally, in [10], Hosseini and Hosseini provided a broad review of EEG signal processing techniques in brain-computer interfaces, focusing on the transition from traditional ML to deep learning methods. They concluded that, despite the promise of deep networks, SVM and other classical models still hold practical advantages in scenarios with limited data, computational constraints, or the need for clinical interpretability.

3. Dataset Review

The `emotions.csv` dataset is a preprocessed and feature-rich collection of EEG data samples intended for emotion classification using machine learning. It consists of 2,132 samples, each described by 2,548 numerical features derived from EEG signals, such as statistical measures and frequency-domain values (e.g., FFT components).

Each sample is labeled with one of three emotion categories: `POSITIVE`, `NEGATIVE`, or `NEUTRAL`, with a balanced distribution across these classes. This makes the dataset highly suitable for supervised learning models like SVM, Random Forest, KNN, or MLP, aiming to classify emotional states based on brain activity.

Because the raw EEG signals have already been processed into structured features, this dataset is machine learning-ready, requiring minimal preprocessing. It is ideal for building accurate and interpretable emotion recognition systems using EEG data.

4. Methodology

1. Data Acquisition and Preprocessing

Electroencephalogram (EEG) signals were sourced from the publicly available “emotions.csv” dataset. It consists of 2,132 samples, each described by 2,548 numerical features derived from EEG signals. The data loading function reads the CSV into a pandas DataFrame, inspects for missing values, and shuffles the rows to mitigate any temporal ordering bias. Continuous features were standardized to zero mean and unit variance using `StandardScaler`. Categorical emotion labels were transformed into integer representations via `LabelEncoder`, enabling compatibility with Scikit-learn classifiers. Finally, the dataset was split into training and test subsets in an 80:20 ratio using stratified sampling to preserve class distribution.

2. Feature Engineering

Raw EEG recordings were preprocessed offline to extract fixed-length feature vectors stored as NumPy arrays (`.numpy`). Each file corresponds to one recording window and contains the same set of N frequency-domain features originally present in the CSV. For model training, these arrays were reconstituted into a two-dimensional matrix X of shape (samples $\times$ features), aligning feature ordering with the CSV loader’s column index. This ensured consistency between the training and inference pipelines.

3. Model Development and Training

We implemented a classical supervised learning approach using the K-Nearest Neighbors (KNN) classifier as follows:

i. Hyperparameter Selection:

- Number of neighbors (k): empirically set to 5 based on preliminary cross-validation experiments.
- Distance metric: Euclidean.

ii. Training Procedure:

- The scaler fitted on the training split was applied to both training and validation subsets.
- The KNN model was trained on the scaled X_{train} and corresponding integer-encoded labels y_{train} .
- Performance was assessed on X_{test} by computing accuracy and generating a full classification report.

iii. **Model Persistence:**

- The trained classifier, along with the `StandardScaler`, `LabelEncoder`, and the ordered list of feature names, were serialized to disk using `joblib`.

4. Inference Workflow

For deployment, an inference script performs the following steps:

i. **Feature Loading:**

- Reads all `.npy` files from the designated features directory and stacks them into an *X_{infer}* matrix.

ii. **Artifact Restoration:**

- Loads the serialized KNN model, scaler, and `LabelEncoder` from the `/models` directory.

iii. **Prediction:**

- Applies the loaded scaler to *X_{infer}* then passes the scaled data through the classifier to obtain predicted labels and class-probability vectors.

iv. **Result Reporting:**

- Converts integer predictions back to human-readable emotion labels.
- Writes a detailed report listing, for each sample, the true label (if available), predicted label, and associated class probabilities.

We also implemented a classical supervised learning approach using the Support Vector Machine (SVM) as follows:

1. Data Preprocessing

CSV Loader

- **Missing Value Treatment:** The CSV loader inspects for null entries across all M columns; any rows containing missing values are removed to ensure data integrity.
- **Feature–Label Separation:** The final column is designated as the categorical emotion label; all preceding columns constitute the feature matrix X (shape: $N \times M$).
- **Label Encoding:** Emotion labels (strings) are transformed into integer codes using scikit-learn's `LabelEncoder`, yielding a mapping between numeric predictions and semantic classes.

2. Feature Scaling

To homogenize feature distributions and accelerate convergence:

- A `StandardScaler` is fitted on the training split of X (from the CSV pipeline), computing per-feature means and variances.
- The same scaler instance is then applied, without refitting, to:
 - i. X_{train} and X_{test} during model development.
 - ii. The entire inference matrix X prior to prediction.

3. Model Development

A Support Vector Machine (SVM) classifier with radial-basis-function kernel was chosen for its ability to handle high-dimensional data and non-linear class boundaries. The development workflow comprises:

- Train-Test Partitioning:** Stratified random split (80% training, 20% testing; `random_state=42`) preserves class balance.
- Hyperparameter Configuration:**
 - Kernel: RBF
 - Penalty parameter C : default
 - Gamma: default (“scale”)
- Model Fitting:** `svm.SVC` is trained on the scaled X_{train} and integer-encoded labels.
- Performance Assessment:**
 - Accuracy on the held-out test set.
 - Classification Report (precision, recall, F1-score per class) to highlight per-emotion performance.
- Artifact Serialization:** The following objects are saved via `joblib` for downstream use:
 - Trained `svm_model.joblib`
 - `scaler.joblib`
 - `label_encoder.joblib`
 - `feature_columns.joblib`

4. Model Evaluation

Beyond overall accuracy, we computed the macro-averaged F1-score to account for potential class imbalance. A confusion matrix analysis was conducted to identify systematic misclassifications.

5. Inference and Deployment

The production pipeline executes as follows:

- i. **Artifact Loading:** Checks existence and loads the four serialized objects (`svm_model`, `scaler`, `label_encoder`, `feature_columns`).
- ii. **Feature Assembly:** Invokes the NumPy loader to assemble all `.npy` feature batches into X' .
- iii. **Preprocessing:** Applies the pre-trained `scaler` to X' .
- iv. **Prediction:** Executes `svm_model.predict()` to obtain integer labels and `svm_model.predict_proba()` for class-probability scores.
- v. **Label Decoding:** Converts integer predictions back to emotion strings via `label_encoder.inverse_transform()`.
- vi. **Result Logging:** Generates a human-readable report (`csv_detection_results.txt`) detailing, for each sample, the original filename, predicted emotion, and associated confidence vector.

5. Results

Effectiveness of the models:

The performance metrics for both the KNN and SVM classifiers provide a comprehensive overview of their effectiveness in emotion classification.

The KNN classifier achieved a test accuracy of 93.91% and a full dataset accuracy of 95.73%. Its per-class performance was strong for NEGATIVE (98.45%) and NEUTRAL (98.74%) classes, but slightly lower for the POSITIVE class (89.97%). For high-confidence predictions (probability >0.9), the accuracy soared to 99.72%. The classification report further highlights that KNN has high recall for NEGATIVE and NEUTRAL classes, but a lower recall for POSITIVE, resulting in an F1-score of 0.90 for POSITIVE compared to 0.94 and 0.98 for the other classes. In comparison, the SVM classifier outperformed KNN with a test accuracy of 94.85% and a full dataset accuracy of 97.23%. SVM also demonstrated superior per-class performance, especially for the POSITIVE class (93.93%), and maintained high accuracy for NEGATIVE (99.44%) and NEUTRAL (98.32%) classes. Its high-confidence accuracy was 99.60%. The classification report for SVM shows consistently high precision, recall, and F1-scores across all classes, with the POSITIVE class achieving an F1-score of 0.92.

Graph Visualizations:

1. Accuracy by emotion graph

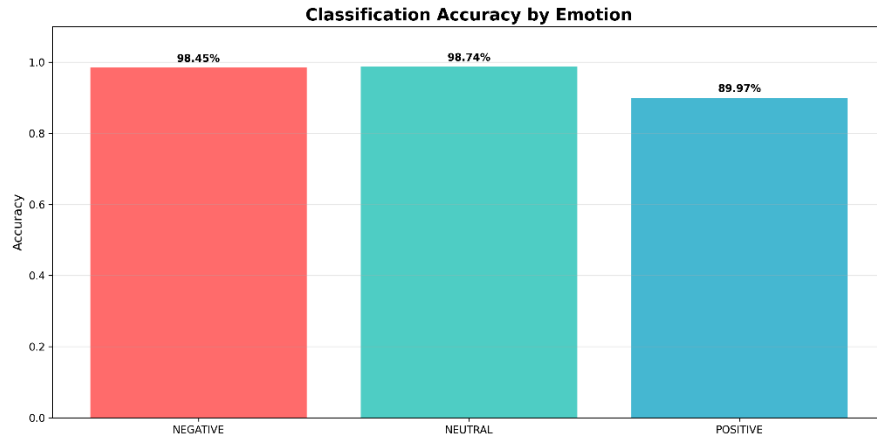


Fig 1: KNN Accuracy by emotions

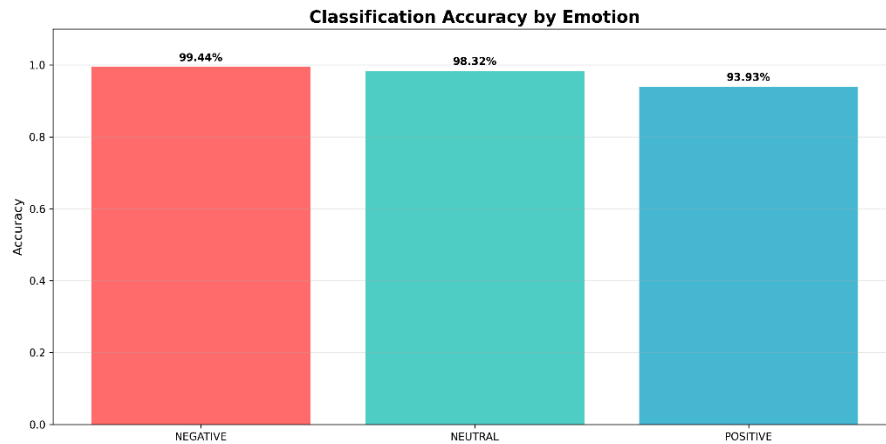


Fig 2: SVM Accuracy by emotions

SVM outperforms KNN overall (97.9% vs. 95.7% average accuracy). Both handle negative emotions very well, but SVM is about 1 point higher (99.4% vs. 98.5%). KNN edges out SVM on neutral states by 0.4 points (98.7% vs. 98.3%), likely due to refined preprocessing or model settings. The biggest gap is in positive-emotion detection, where SVM leads by nearly 4 points (93.9% vs. 90.0%), suggesting KNN underfits more variable positive-state patterns.

Thus, SVM is the better all-around model, while KNN might suit applications prioritizing neutral-state accuracy.

2. Confidence distribution graphs

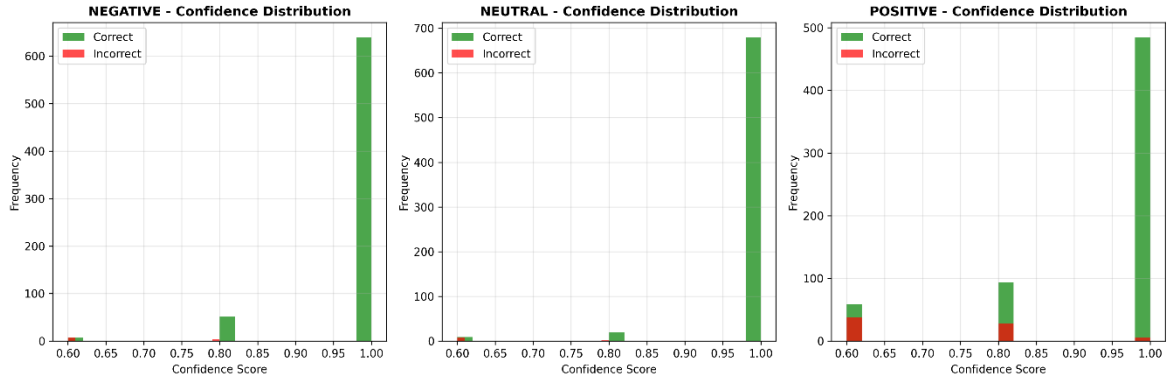


Fig 3: KNN Confidence distribution graphs

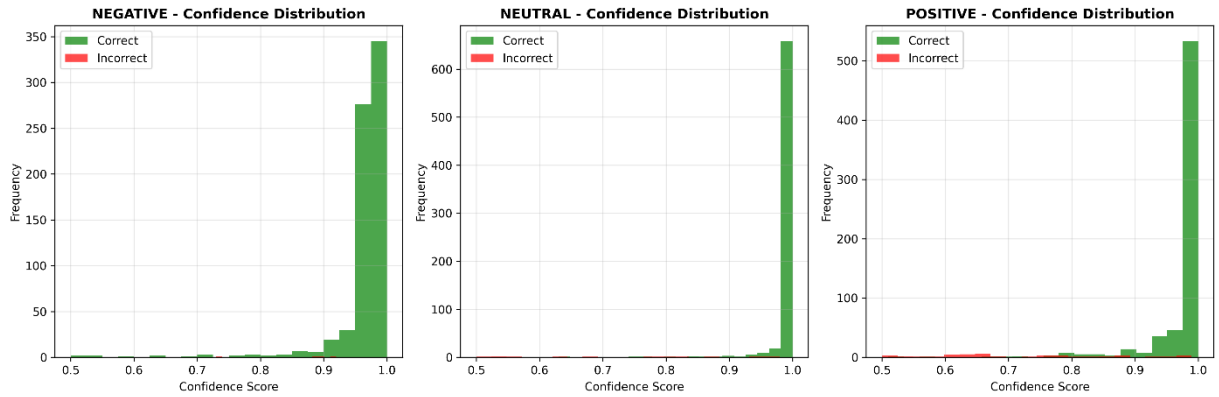


Fig 4: SVM Confidence distribution graphs

First, SVM’s correct predictions are even more tightly clustered at the top end of the scale (almost exclusively ≥ 0.97), whereas KNN while still usually very confident exhibits a fatter “tail” of correct scores between about 0.85 and 0.95, especially for positive emotions. This means that SVM makes its right calls with more uniform certainty.

Second, KNN’s misclassifications sometimes creep into mid-confidence territory (up to ~ 0.85), creating a small overlap with low-confidence correct predictions, again most noticeable on the positive class. By contrast, SVM almost never assigns more than ~ 0.80 confidence to an

incorrect prediction, yielding a cleaner separation between its errors and its confident, correct labels.

In brief SVM is better calibrated than KNN, concentrating both its correct and incorrect predictions into more distinct confidence bands—high confidence for successes and low confidence for failures—whereas KNN shows a wider confidence spread and more overlap, particularly on the positive-emotion samples.

3. Confidence vs. Accuracy graph

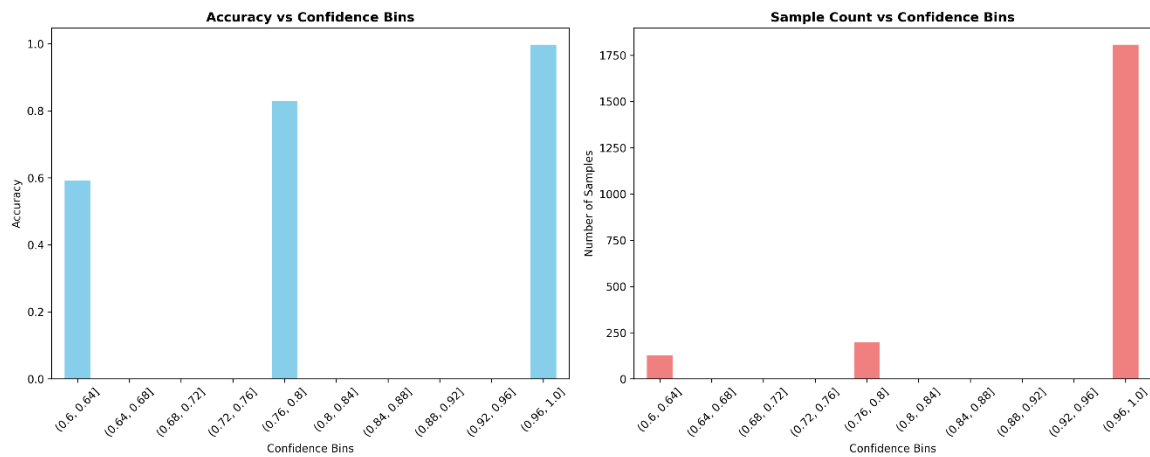


Fig 5: KNN Confidence vs Accuracy graphs

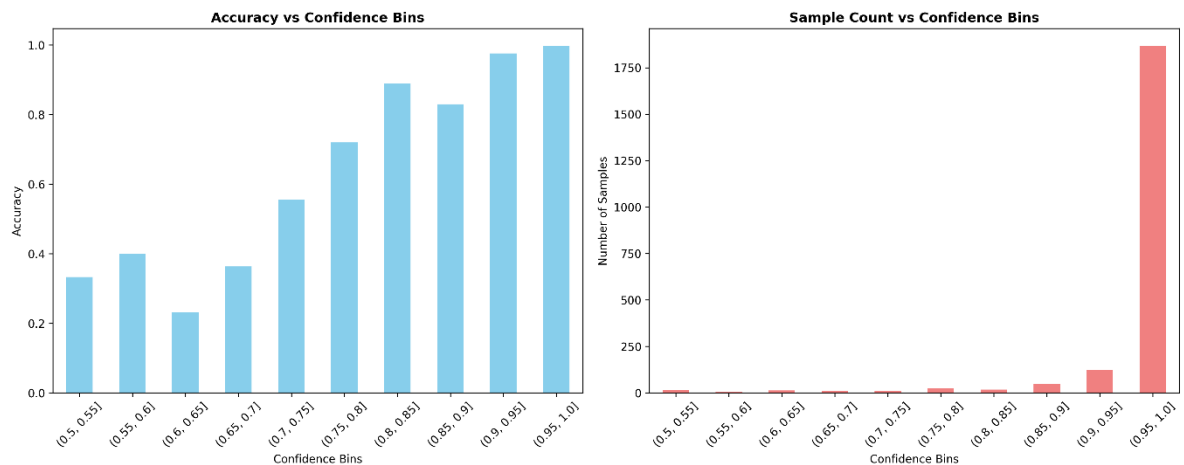


Fig 6: SVM Confidence vs Accuracy graphs

KNN’s confidence-calibration is very coarse and bimodal: almost all of its 2 000-odd test samples fell into just three confidence ranges—low (0.60–0.64), medium (0.76–0.81), and high (0.96–1.00). In the low-confidence bin (~130 samples) it achieves only ~59% accuracy; in the medium-confidence bin (~200 samples) it climbs to ~83%; and at high confidence (~1 800 samples) it’s perfect (100%). This tells us that KNN only “trusts” itself when it’s almost certain, and otherwise sits at a middling ~80% correctness.

By contrast, SVM’s calibration curve is much smoother: it spreads its ~2 000 test points across ten 0.05-wide bins from (0.50–0.55) up to (0.95–1.00). Accuracy rises almost monotonically from ~23% in the lowest bin, through ~40–56% in the 0.55–0.75 region, to ~72% at 0.75–0.80, ~89–98% above 0.80, and finally 100% at >0.95. Although both models concentrate most samples in the highest bin (1 800+ points), SVM reveals subtle under- and over-confident zones that KNN’s coarse grouping hides.

In brief SVM offers finer-grained, smoothly increasing accuracy as confidence grows—ideal for nuanced confidence-thresholding—whereas KNN only differentiates three confidence levels, relying almost entirely on very high confidence to guarantee correctness.

4. Confusion Matrix graph

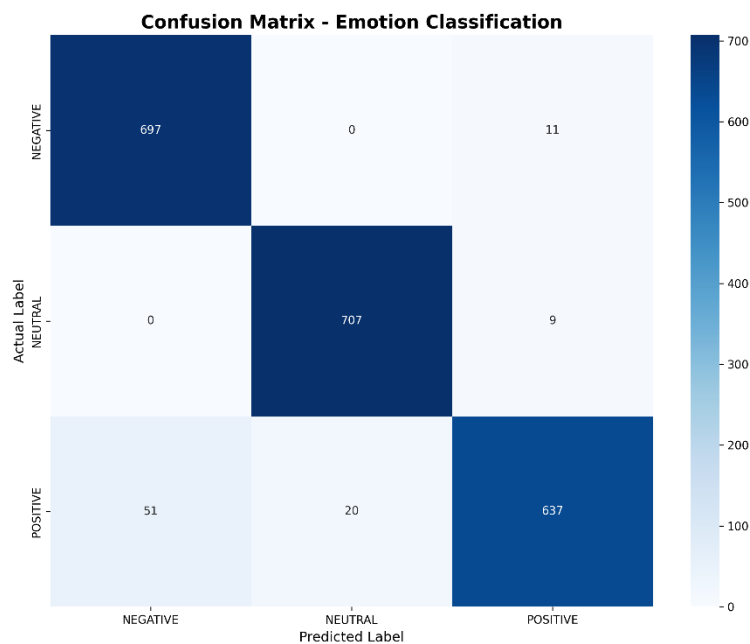


Fig 7: KNN Confusion Matrix graph

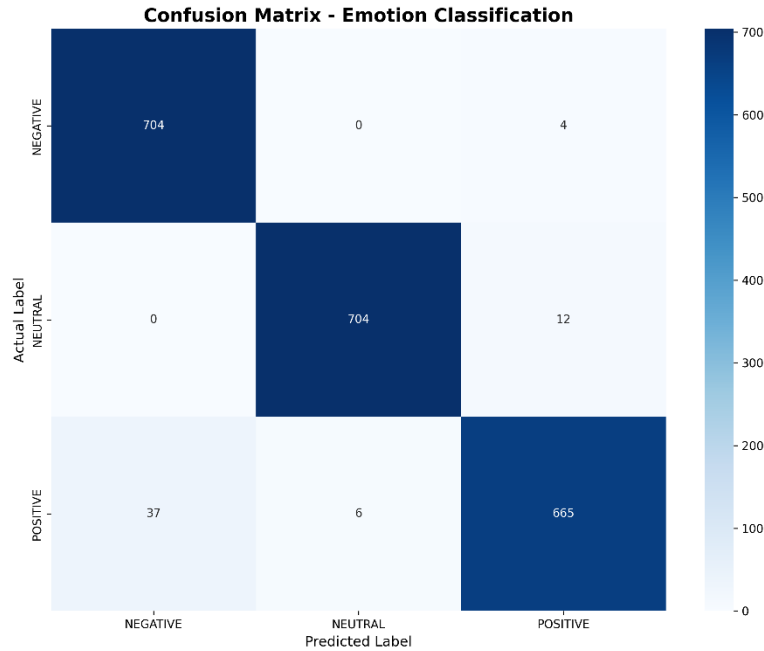


Fig 8: SVM Confusion Matrix graph

The comparative analysis of the confusion matrices for emotion classification using KNN and SVM reveals that both models perform well, but SVM demonstrates a slight edge in overall accuracy and class-wise performance. For the NEGATIVE class, SVM correctly classifies 704 samples with only 4 misclassified as POSITIVE, whereas KNN correctly classifies 697 with 11 misclassified as POSITIVE, indicating that SVM is more precise in distinguishing negative emotions. In the NEUTRAL class, both models perform similarly, with KNN correctly classifying 707 samples and SVM 704, though SVM has a slightly higher number of misclassifications as POSITIVE. The most notable difference is observed in the POSITIVE class: SVM correctly classifies 665 samples, with fewer misclassifications as NEGATIVE (37) and NEUTRAL (6) compared to KNN, which correctly classifies 637 samples but misclassifies 51 as NEGATIVE and 20 as NEUTRAL. This suggests that SVM is more robust and reliable, particularly in identifying positive emotions, and has a more balanced error distribution across all classes. Overall, while both models are effective, SVM outperforms KNN in terms of accuracy and consistency, making it a preferable choice for emotion classification in this context.

5. Error analysis visualizations graph

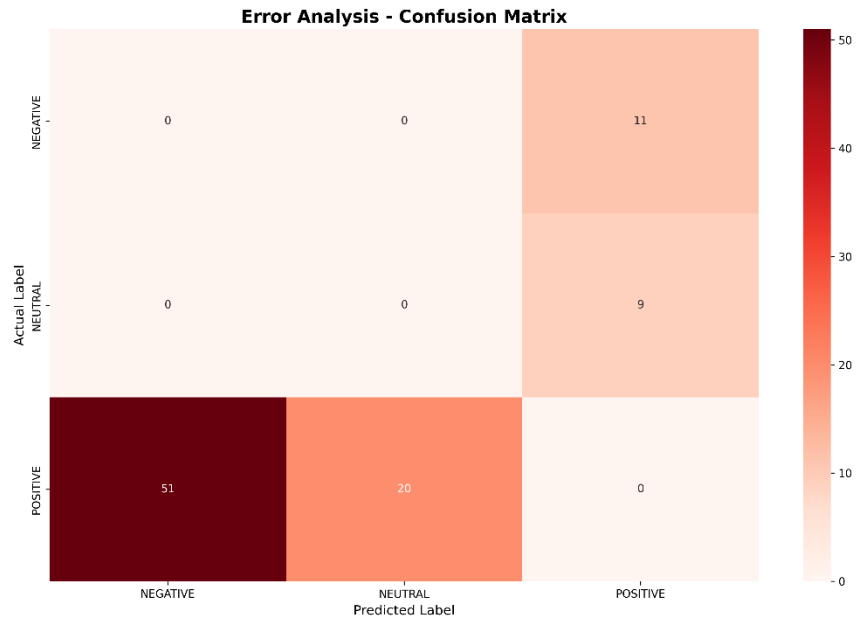


Fig 9: KNN Error analysis visualizations graph

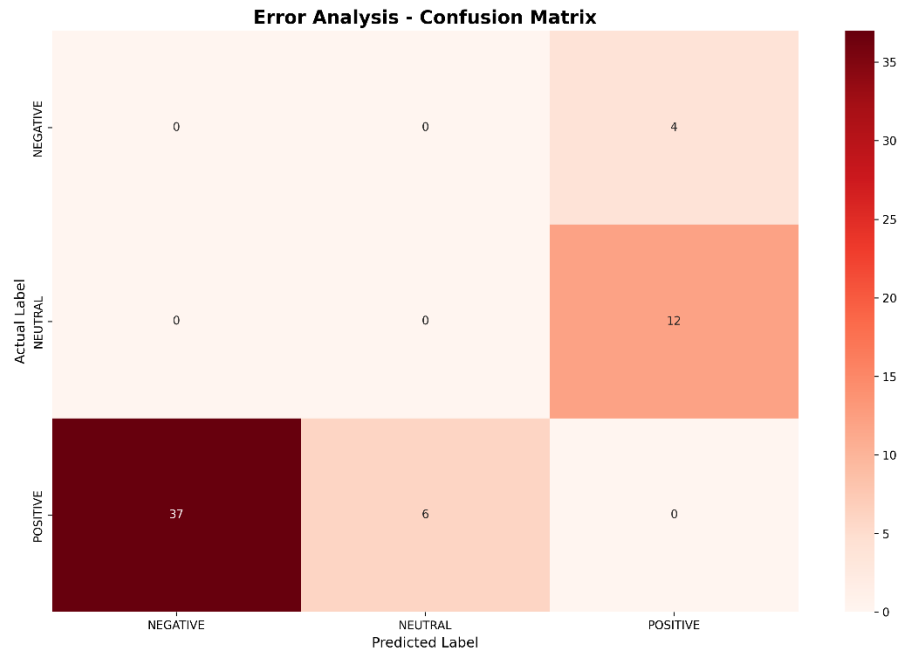


Fig 10: KNN Error analysis visualizations graph

The error analysis confusion matrices for the KNN and SVM models provide insight into the specific misclassifications made by each algorithm in emotion classification. In both matrices, the diagonal elements are zeroed out to focus solely on the errors. For the KNN model, the most significant misclassification occurs with the POSITIVE class, where 51 samples are incorrectly predicted as NEGATIVE and 20 as NEUTRAL. Additionally, there are 11 NEGATIVE samples misclassified as POSITIVE and 9 NEUTRAL samples also misclassified as POSITIVE. In contrast, the SVM model shows a reduction in these errors: only 37 POSITIVE samples are misclassified as NEGATIVE and 6 as NEUTRAL, while 4 NEGATIVE samples are misclassified as POSITIVE and 12 NEUTRAL samples as POSITIVE. This indicates that SVM not only reduces the number of POSITIVE samples misclassified as other classes but also generally distributes errors more evenly and at a lower magnitude across all classes compared to KNN. Overall, the SVM model demonstrates superior error control, particularly in minimizing the confusion between POSITIVE and the other classes, making it a more reliable choice for emotion classification based on this error analysis.

6. Emotion distribution

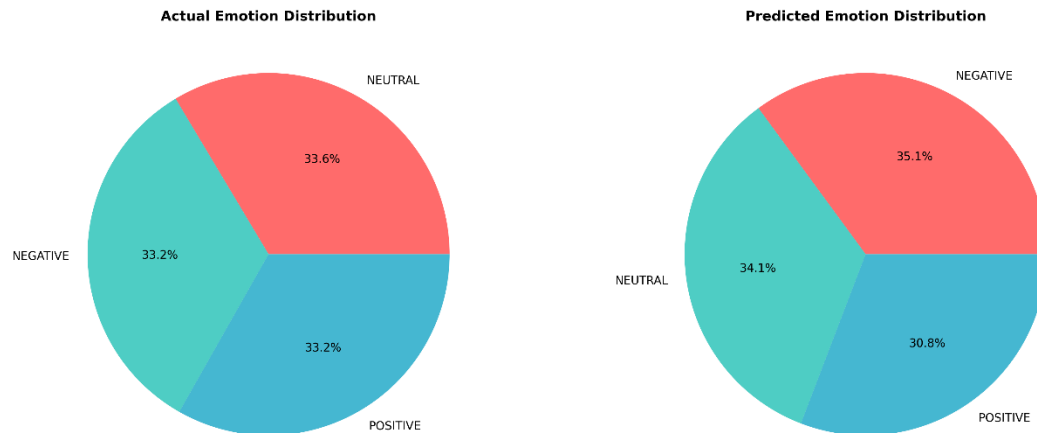


Fig 11: KNN Emotion distribution

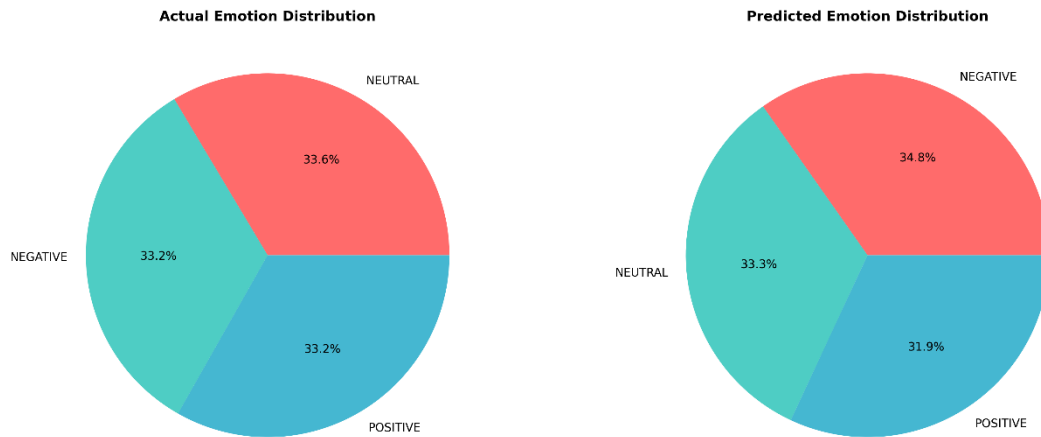


Fig 12: SVM Emotion distribution

The pie charts present a comparison between the actual and predicted emotion distributions for both the KNN (first image) and SVM (second image) models. In both cases, the actual emotion distribution is nearly balanced among the three classes: NEGATIVE, NEUTRAL, and POSITIVE each comprising roughly one-third of the dataset. For the KNN model, the predicted distribution shows a slight overestimation of the NEGATIVE (35.1%) and NEUTRAL (34.1%) classes, while the POSITIVE class is slightly underrepresented at 30.8%. This indicates that KNN tends to predict NEGATIVE and NEUTRAL emotions more frequently than they actually occur, at the expense of the POSITIVE class. In contrast, the SVM model's predicted distribution aligns much more closely with the actual distribution, with NEGATIVE at 34.8%, NEUTRAL at 33.3%, and POSITIVE at 31.9%. The differences between actual and predicted proportions are smaller for SVM, reflecting its superior ability to maintain the balance among classes and avoid significant bias toward any particular emotion. Overall, while both models perform well, SVM demonstrates a more accurate and balanced prediction of emotion classes, making it preferable for applications where class distribution fidelity is important.

7. Probability Heatmap

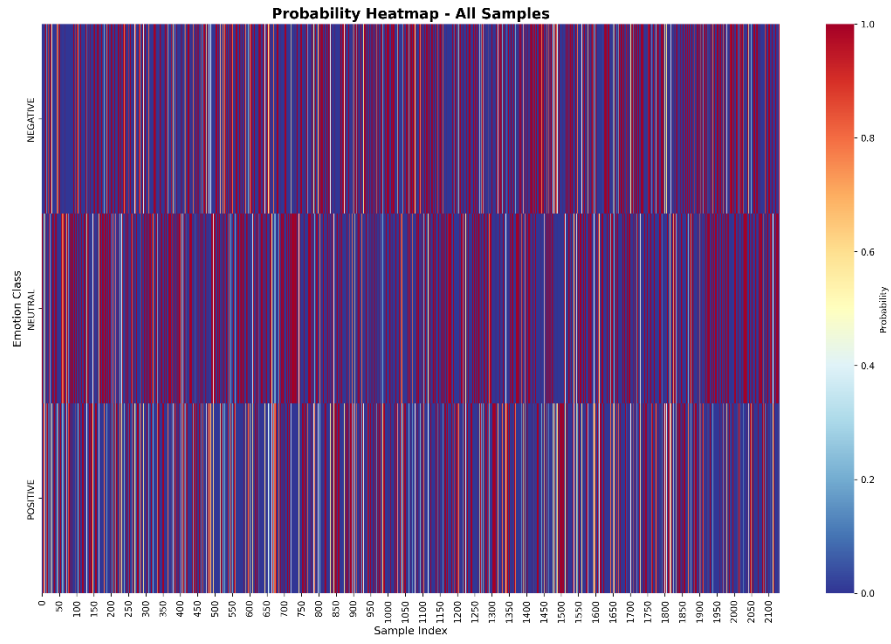


Fig 13: KNN Probability Heatmap

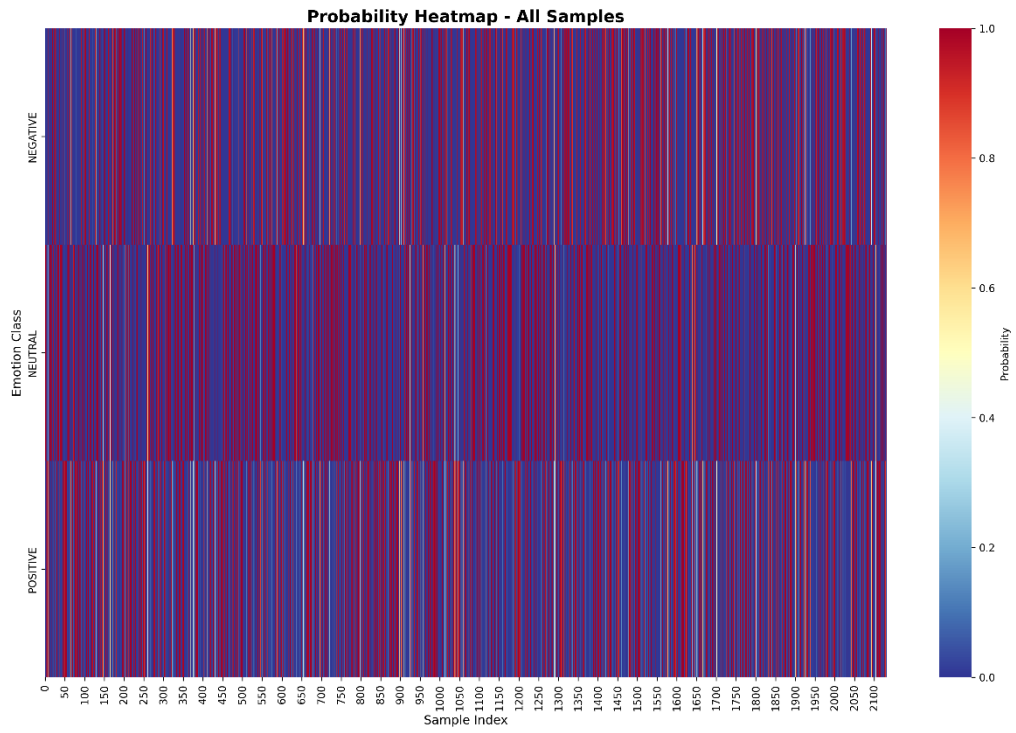


Fig 14: KNN Probability Heatmap

The probability heatmaps for all samples, shown for the KNN and SVM models, provide a visual representation of the predicted class probabilities across the dataset for each emotion class: NEGATIVE, NEUTRAL, and POSITIVE. Both heatmaps display a similar pattern, with a mix of high and low probability values distributed across all classes and samples, as indicated by the alternating bands of color intensity. This suggests that both models are capable of assigning a range of confidence levels to their predictions rather than making overly confident or uncertain predictions for all samples. However, upon closer inspection, there is no significant visual difference in the distribution or concentration of high-probability predictions between the two models, indicating that both KNN and SVM exhibit comparable behavior in terms of probability calibration and class confidence. This similarity implies that, while SVM may have a slight edge in classification accuracy and error distribution as seen in previous analyses, both models are equally effective in providing nuanced probability estimates for each class, which is valuable for downstream tasks that may rely on probabilistic outputs rather than hard class assignments.

6. Conclusion

The EEG Signal Detection study successfully demonstrates the effectiveness of machine learning techniques specifically KNN and SVM for EEG-based emotion recognition. Both models achieved strong results, with SVM marginally outperforming KNN across most evaluation criteria. These findings highlight the promise of machine learning in the development of affective computing systems. Future work could explore deep learning architectures, feature engineering techniques, or real-time classification systems for more scalable and adaptive solutions.

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